UNIT – II

Green Building Concepts and Practices

Indian Green Building Council, Green Building Moment in India, Benefits Experienced in Green Buildings, Launch of Green Building Rating Systems, Residential Sector, Market Transformation; Green Building Opportunities and Benefits: Opportunities of Green Building, Green Building Features, Material and Resources, Water Efficiency, Optimum Energy Efficiency, Typical Energy Saving Approach in Buildings, LEED India Rating System and Energy Efficiency

2.1 Indian Green Building Council (IGBC)

2.1.1 Green Building Moment in India

The housing sector in India is growing at a rapid pace and contributing immensely to the growth of the economy. This augurs well for the country and now there is an imminent need to introduce green concepts and techniques in this sector, which can aid growth in a sustainable manner.

Green concepts and techniques in the residential sector can help address national issues like water efficiency, energy efficiency, and reduction in fossil fuel use in commuting, handling of consumer waste and conserving natural resources. Most importantly, these concepts can enhance occupant health, happiness and well-being.

Against this background, the Indian Green Building Council (IGBC) has launched 'IGBC Green Homes® Rating System' to address the National priorities. By applying IGBC Green Homes® criteria, homes which are sustainable over the life cycle of the building can be constructed. This rating programme is a tool which enables the designer to apply green concepts and criteria, so as to reduce the environmental impacts, which are measurable. The programme covers methodologies to cover diverse climatic zones and changing lifestyles.

IGBC Green Homes® is the first rating programme developed in India, exclusively for the residential sector. It is based on accepted energy and environmental principles and strikes a balance between known established practices and emerging concepts. The system is designed to be comprehensive in scope, yet simple in operation.

2.1.2 Benefits of Green Homes

Green homes can have tremendous benefits, both tangible and intangible.

1. The most tangible benefits are
   - the reduction in water and energy consumption right from day one of occupancy.
   - the energy savings could range from 20 - 30 % and water savings around 30 - 50%.
2. Intangible benefits of green homes include:
   - enhanced air quality,
   - excellent daylighting,
   - health & well-being of the residents,
   - safety benefits and

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- conservation of scarce national resources.

2.1.3 Launch of Green Building Rating Systems- Residential Sector

2.1.3.1 Indian Green Building Council (IGBC)

IGBC has set up the Green Homes Core Committee to focus on residential sector. The committee includes builders, developers, home owners, architects, consultants, experts on building science, manufacturers and industry representatives. The varied experience and professions of the committee members bring in a holistic perspective in the process of developing the rating programme.

Evolution of Rating System-IGBC

IGBC, in its endeavor to extend green building concepts to all building types envisioned a rating programme exclusively for homes.

- A core committee was formed under the leadership of Ar Sharukh Mistry, Mistry Architects, Bangalore.
- The pilot version of the programme was launched in July 2008. The rating programme has evoked tremendous response from the stakeholders.
- Based on the feedback and learning from various projects that have implemented the rating programme, the latest version 2.0 has been launched in March 2012. The rating system is designed to suit Indian climate and construction practices.

About 220 members representing 120 organisations have participated in the development of the rating programme. As on date (March 2012), over 400 million sq.ft. of built-up area is registered under the rating programme.

The rating system will be subject to review by the core committee, every year, to ensure that it is updated and contemporary.

- IGBC Green Homes Rating System is a voluntary and consensus-based programme.
- The rating system has been developed based on materials and technologies that are presently available.
- The objective of IGBC Green Homes Rating is to facilitate the creation of water efficiency, handling of house-hold waste, energy efficiency, healthy, comfortable and environmentally-friendly houses.
- IGBC Green Homes Rating System is a measurement system designed for rating new and major, renovated residential buildings which are broadly classified into two construction types:
  1. Individual residential unit
  2. Multi-dwelling residential units
     - ✓ Gated communities.
     - ✓ High-rise residential apartments.
     - ✓ Hostels, Service apartments, Resorts, Motels and Guest houses.

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The rating programme will also undergo periodic revisions to incorporate the latest advances and changes.

IGBC Green Homes® Rating System

IGBC Green Homes Rating system addresses green features under the following categories:

1. Site Selection and Planning
2. Water Efficiency
3. Energy Efficiency
4. Materials & Resources
5. Indoor Environmental Quality
6. Innovation & Design Process

The guidelines detailed under each mandatory requirement & credit enables the design and construction of green homes of all sizes and types. Different levels of green building certification are awarded based on the total credits earned. However, every green home should meet certain mandatory requirements, which are non-negotiable.

The various levels of rating awarded are:

Certification Level	Recognition
Certified	Best Practices
Silver	Outstanding Performance
Gold	National Excellence
Platinum	Global Leadership

2.1.3.2 GRIHA - Green Rating for Integrated Habitat Assessment

- Human Habitats (buildings) interact with the environment in various ways. Throughout their life cycles, from construction to operation and then demolition, they consume resources in the form of energy, water, materials, etc. and emit wastes either directly in the form of municipal wastes or indirectly as emissions from electricity generation.
- GRIHA attempts to minimize a building's resource consumption, waste generation, and overall ecological impact to within certain nationally acceptable limits / benchmarks.
- GRIHA is a rating tool that helps people assesses the performance of their building against certain nationally acceptable benchmarks.

The basic Features of GRIHA

A building is assessed based on its predicted performance over its entire life cycle – inception through operation. The stages of the life cycle that have been identified for evaluation are:

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1. Pre-construction stage: intra- and inter-site issues like proximity to public transport, type of soil, kind of land, where the property is located, the flora and fauna on the land before construction activity starts, the natural landscape and land features.

2. Building planning and construction stages: issues of resource conservation and reduction in resource demand, resource utilization efficiency, resource recovery and reuse, and provisions for occupant health and well-being. The prime resources that are considered in this section are land, water, energy, air, and green cover.

3. Building operation and maintenance stage: issues of operation and maintenance of building systems and processes, monitoring and recording of energy consumption, and occupant health and well-being, and also issues that affect the global and local environment.

GRIHA – Rating Variants

1. *SVAGRIHA Pre-Certification*

    - SVAGRIHA Precertification is applicable only for projects which are less than 2500 sq m built-up area. Any building, except for a factory building, is accepted under the SVAGRIHA Precertification rating system

2. *SVAGRIHA*

    - SVAGRIHA can help in design and evaluation of individual residences, small offices, schools, motels, commercial buildings etc

3. *GRIHA Rating*

    - All new construction projects with built up area more than 2500 m^2 (excluding parking, basement area, and typical buildings) are eligible for pre-certification

4. *GRIHA AH Precertification*

    - All projects that are in-line with the PMAY requirements will be eligible for GRIHA AH since the rating strives to integrate sustainability in the residences in line with the PMAY.
    - All the upcoming projects that have approval/sanction letter issued by government agency (Central/State) confirming that the project is being developed as per Pradhan Mantri Awas Yojana (PMAY) are eligible for GRIHA AH rating.

5. *GRIHA for Existing Building Rating*

    - All operational buildings having a built-up area greater than 2,500 sq. m. are eligible for certification under GRIHA for Existing Buildings rating. The typology of buildings include: offices, retail spaces, institutional buildings, hotels, hospital buildings, healthcare facilities, residences, and multi-family high-rise buildings.

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6. *GRIHA for Existing Day Schools*

    - All day schools (excluding hostel/ residential facilities) which have been operational for at least 1 year are eligible for certification under GRIHA for Existing Day Schools rating.

7. *GRIHA for Cities*

    - The GRIHA for CITIES rating has been structured as a framework for sustainable development of a city, to be achieved by measuring 'greenness' of existing as well as proposed cities.
    - The rating targets all cities, towns with civic/local bodies, such as Municipal Corporation and Municipal Council.

8. GRIHA for Large Developments

    - The intent of GRIHA LD is to provide a consolidated framework for assessment of environmental impacts of large-scale developments.

GRIHA – Benefits

This benefits the community at large with the improvement in the environment by reducing GHG (greenhouse gas) emissions, reducing energy consumption and the stress on natural resources.

Some of the benefits of a green design to a building owner, user, and the society as a whole are as follows:

1. Reduced energy consumption without sacrificing the comfort levels
2. Reduced destruction of natural areas, habitats, and biodiversity, and reduced soil loss from erosion etc.
3. Reduced air and water pollution (with direct health benefits)
4. Reduced water consumption
5. Limited waste generation due to recycling and reuse
6. Reduced pollution loads
7. Increased user productivity
8. Enhanced image and marketability

2.2 Role of Green Building Rating Systems in Market Transformation:

The fundamental market transformation can be seen across key areas:

1. The Cost Premium & Feasibility

- Before Green Rating Systems: Building green was an expensive, experimental luxury. Because sustainable technologies were rare and unstandardized, developers faced a 10% to 15% upfront capital cost premium. Most eco-friendly technologies, specialized

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glass, or high-efficiency HVAC units had to be imported, making sustainability economically unviable for mainstream developers.

- After Green Rating Systems: The upfront cost premium has collapsed to a modest 1% to 3.6% for standard projects, with many developers achieving cost parity. The standardized rating points created massive demand, which drove local manufacturing economies of scale. These upfront costs are now easily recovered within 2 to 3 years through operational utility savings.

2. The Materials & Construction Supply Chain

- Before Green Rating Systems: The supply chain was highly fragmented. If a builder wanted "eco-friendly" options, they relied on unverified vendor claims. Basic items like traditional clay bricks (which destroy topsoil) and standard OPC cement were the absolute default.
- After Green Rating Systems: India now features a multibillion-dollar localized green building materials market. Rating systems forced structural accountability. Industrial waste byproducts like fly ash and slag are now standard components in concrete blends. Products like Autoclaved Aerated Concrete (AAC) blocks, low-VOC paints, and high-performance glass are locally manufactured commodities. Programs like CII's *GreenPro* provide formal labels so builders can instantly source certified eco-friendly materials.

3. Commercial Real Estate Demand & Tenant Behavior

- Before Green Rating Systems: Location, square footage, and exterior aesthetics were the only metrics that dictated "Grade A" office spaces. Energy and water efficiency were completely ignored during leasing negotiations because tenants viewed utility bills as an inevitable cost of doing business.
- After Green Rating Systems: Sustainability is a core corporate real estate requirement. With multinational corporations and Global Capability Centers (GCCs) bound by international ESG (Environmental, Social, and Governance) targets, the vast majority of India's Grade A office stocks are green-certified. Non-certified commercial properties now face a "brown discount"—meaning they suffer from higher vacancy rates and lower rental yields because modern corporate tenants actively filter them out.

4. Architectural Design & Operational Data

- Before Green Rating Systems: "Green washing" ran rampant. A developer could plant a few trees or put glass on a building and market it as eco-friendly. Buildings were treated as static structures; sub-metering was rare, and real-time energy or water tracking was practically non-existent.
- After Green Rating Systems: Sustainability is now completely data-driven. To achieve certification, projects must verify true structural performance through smart sub-metering, Building Management Systems (BMS), and continuous Indoor Air Quality

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(IAQ) monitoring. The market shifted from subjective marketing claims to verified, empirical metrics: a certified building guarantees 20% to 30% lower energy use and 30% to 50% lower water consumption.

5. Public Policy & Government Regulations

- Before Green Rating Systems: Government building regulations (municipal bye-laws) were basic and focused entirely on setbacks, safety, and height restrictions. Environmental clearances were tedious, slow, and offered no rewards for sustainable design.
- After Green Rating Systems: Rating frameworks are deeply woven into public policy. They heavily influenced national guidelines like the Energy Conservation and Sustainable Building Code (ECSBC). Furthermore, state governments and municipal corporations actively incentivize these ratings. Today, achieving a high IGBC or GRIHA rating serves as a regulatory fast-track, granting developers concrete financial perks like 1% to 5% additional Floor Area Ratio (FAR) free of cost, property tax rebates, or streamlined environmental clearances.

2.3 Green Building Opportunities

Green buildings offer numerous opportunities for improving environmental sustainability, economic efficiency, and occupant well-being. They encourage the adoption of innovative technologies and resource-efficient practices throughout the building life cycle.

1. The Developer & Investor (The Business Case)

For those who finance and own real estate, green buildings have evolved from a "corporate social responsibility" box to check into a powerful tool for maximizing profits and protecting wealth.

- *Unlocking Global Capital:* International investors and major funds now operate under strict ESG (Environmental, Social, and Governance) rules. Certified green properties function like an elite passport, granting access to foreign direct investment (FDI) and low-interest "green loans" that aren't available to standard projects.
- *Higher Yields, Lower Risk:* High-profile corporate tenants are willing to pay a premium for certified Gold or Platinum spaces to meet their own carbon-reduction targets. These buildings fill up faster, command higher rent, and experience much lower tenant turnover.
- *Avoiding Asset Obsolescence:* As environmental regulations tighten, traditional buildings face a "brown discount"—they become more expensive to run and harder to lease. Going green future-proofs the property’s long-term market value.

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2. The Architect, Civil Engineer, & Consultant (The Technical Edge)

For design and construction professionals, the green building shift moves engineering away from standard, repetitive practices and pushes it toward high-value, data-driven innovation.

- *The Rise of Smart Tech:* Earning top rating points requires embedding cutting-edge technology directly into the building's skeleton. This opens massive opportunities in:
    - *BIM (Building Information Modelling) & Digital Twins:* Using advanced software to simulate daylight, airflow, and energy use before a single brick is laid.
    - *AI & IoT Integration:* Designing automated systems where smart sensors adjust cooling, lighting, and ventilation based on real-time room occupancy and Carbon di oxide levels.
- *High-Demand Career Niches:* Navigating complex rating systems has created a lucrative job market for specialized professionals. There is a soaring demand for Certified Energy Auditors, Life Cycle Assessment (LCA) experts, and Accredited Professionals (like IGBC AP or GRIHA CP) who manage the certification process.

3. The Manufacturer & Materials Vendor (The Industrial Boom)

For companies that produce and supply building products, rating systems have completely rewritten the rules of what materials a job site requires.

- *Massive Green Supply Chain Market:* Rating credits heavily reward projects that buy local, recycled, and eco-certified materials. This has opened up a multibillion-dollar domestic market for manufacturers producing items like:
    - High-performance low-E glass and advanced insulation.
    - Eco-friendly walling like Autoclaved Aerated Concrete (AAC) blocks and fly-ash bricks.
    - Low-VOC paints, sustainable adhesives, and recycled-content flooring.
- *Water & Waste Tech Innovation:* Because rating systems demand drastic reductions in water usage, there is an absolute boom for companies selling localized Sewage Treatment Plants (STPs), smart greywater recycling systems, and ultra-efficient plumbing hardware.

4. Policy Makers & Municipalities (The Urban Strategy)

For local and state governments, green building frameworks provide a pre-made toolkit to upgrade city infrastructure without breaking the public budget.

- *Leveraging Public Incentives:* Governments can drive massive environmental changes simply by offering structural perks instead of direct cash hand-outs. Today, developers targeting high ratings can secure:
    - Extra Floor Area Ratio (FAR): Getting 1% to 5% extra buildable space for free, this instantly increases their saleable square footage.

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    - Fast-Track Approvals: Bypassing slow municipal bottlenecks and moving through environmental clearances at high speed.
- *Relieving Strain on City Infrastructure:* By rewarding on-site solar energy production, rainwater harvesting, and waste recycling, green buildings act as self-sustaining pockets. This drastically reduces the daily load on city power grids, municipal water lines, and crowded landfills.

2.4 Green Building Features

2.4.1 Material and Resources

1. *Separation of House-hold Waste*
    - *IGBC Intent:* Facilitate segregation of house-hold waste at source so as to prevent such waste being sent to land-fills.
    - *Approach and Methodologies:*
      Allocate suitable space on-site for sorting out dry and wet waste. Examine the scope for recycling of waste collected from the project. Identify local dealers to collect and dispose waste material such as paper, plastic, metals, glass, cardboard, organic waste, batteries, E-waste and lamps. Educate residents about various recycling methods.
2. *Organic Waste Management, Post-occupancy*
    - *IGBC Intent:* Ensure effective organic waste management, post-occupancy, so as to prevent waste being sent to land-fills.
    - *Approach and Methodologies:* Calculate approximate quantity of organic waste generated in the building. Design the organic waste treatment plant with suitable capacity to efficiently manage the waste. Typical technologies recognised include digesters, gasifiers, vermi-composting, etc.,
3. *Handling of Construction Waste Materials*
    - *IGBC Intent:* Encourage practices to manage construction waste, thereby, avoiding waste being sent to land-fills.
    - *Approach and Methodologies:* Collect all construction debris generated on-site. Segregate this waste based on their utility. Examine means of reusing such waste within the project or other projects, (or) identify appropriate vendors to divert such waste to manufacturing units which would use them as raw materials. Typical construction debris in residential projects include broken bricks, steel bars, broken tiles, glass, wood waste, paint cans, cement bags, packing materials, etc.
4. *Reuse of Salvaged Materials*
    - *IGBC Intent:* Encourage the use of salvaged building materials and products to reduce the demand for virgin materials, thereby, minimising the impacts associated with extraction and processing of virgin materials.
    - *Approach and Methodologies:* Identify opportunities to incorporate salvaged materials in the building(s) / campus. Consider using salvaged materials for applications such as bricks, doors, flooring, frames, furniture, panelling, etc.,

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5. *Materials with Recycled Content*
   - *IGBC Intent:* Encourage use of materials which contain recycled content to reduce environmental impacts associated with the use of virgin materials.
   - *Approach and Methodologies:* Survey the materials with recycled content identify such local suppliers. Materials with recycled content include aluminium, cement, composite wood, concrete, glass, gypsum, false ceiling, fly ash blocks, steel, tiles, etc.,
6. *Local Materials*
   - *IGBC Intent:* Encourage use of building materials available locally, thereby, minimising the associated environmental impacts resulting from transportation.
   - *Approach and Methodologies:* Survey and identify building materials which are in the specified radius, in early stages of project design. While selecting local materials, ensure that they perform better in terms of strength, maintenance and durability.
7. *Rapidly Renewable Building Materials & Certified Wood*
   - *IGBC Intent:* Minimise use of virgin wood, thereby, encouraging responsible forest management and maximise use of materials which are rapidly renewable.
   - *Approach and Methodologies:* Identify all wood-based applications in the project. Determine the types of materials/ products needed (e.g., doors, windows, furniture, flooring etc.,). Explore the possibility of using rapidly renewable materials and wood certified by Forest Stewardship Council (FSC) or Programme for the Endorsement for Forest Certification (PEFC) or equivalent, for all such wood-based applications. Also, while sourcing wood for various applications, specify the quality or grade of wood required. Locate local suppliers so as to reduce additional costs and environmental impacts caused during transportation.

2.4.2 Water Efficiency

1. *Rainwater Harvesting, Roof & Non-roof*
   - *IGBC Intent:* Enhance ground water table and reduce municipal water demand through effective rain water management.
   - *IGBC Approach and Methodologies:* Survey the water table in the project's location. Design appropriate harvesting system based on the sub surface characteristics. Factors to be considered include weathering, fractures & joints for rocky sites and thickness of aquifer for sedimentary sites.

     Capture rainwater from roof top for reuse. The design should also include flushing arrangement to let out impurities in the first few showers. Such pollutants and impurities include paper waste, leaves, bird droppings, dust, etc.,
2. *Water Efficient Plumbing Fixtures*
   - *IGBC Intent:* Minimise the use of municipal water and reduce load on waste water systems.
   - *Approach and Methodologies:* While selecting water fixtures, look for their efficiencies. The product catalogue or the brochure may detail the flow rates at various pressures. Fixtures are available with ultra-high efficiency which can reduce substantial quantity of water consumption.

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Consider reuse of treated waste water and captured rain water for flushing. Ensure periodic testing of the treated water to meet the quality standards for flushing, as prescribed by Central / State Pollution Control Board.

3. *Landscape Design*
   - *IGBC Intent:* Design landscape to ensure minimum water consumption.
   - *Approach and Methodologies:* During landscape design minimise turf to the extent possible. Select ground covers, shrubs and trees which consume less water. Select species that are well-adapted to the site and which are drought tolerant. Consider xeriscaping as an approach for landscaping.
4. *Management of Irrigation Systems*
   - *IGBC Intent:* Reduce water demand for irrigation through water efficient management systems and techniques.
   - *Approach and Methodologies:* Design irrigation management system based on the requirements of the landscape plan. Conduct a market survey on the technologies available to manage irrigation efficiently. The designer and the installer must work together and ensure the planned performance of the system. Also, identify local manufacturers supplying systems such as moisture sensors, time-based controllers, etc.
5. *Waste Water Treatment and Reuse*
   - *IGBC Intent:* Reduce consumption of potable water and waste water generation to minimise the burden on municipal streams.
   - Approach and Methodologies: Calculate the waste water volumes generated in the building. Design appropriately the capacity of the on-site waste water treatment system. While designing the treatment system, ensure that the treated waste water meets the required quality standards based on its purpose of application. Ensure that the quality of the treated waste water is fit and safe for reuse.

     Prioritise the use of treated waste water such that irrigation, flushing, etc., Excess treated waste water can also be used for make-up water for air-conditioning systems and other purposes. Ensure periodic testing of the treated water to meet the quality standards as prescribed by Central / State Pollution Control Board. Have adequate signages all around the building to caution residents and housekeeping staff that this water is not potable.
6. *Water Metering*
   - *IGBC Intent:* Encourage continuous monitoring to enhance the water performance in the project.
   - *Approach and Methodologies:* Identify all the major water consuming areas and install systems to monitor their consumptions. Develop and implement a measurement and verification mechanism to compare predicted water savings to actual water consumption.

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2.4.3 Optimum Energy Efficiency

1. *CFC-free Equipment*
   - *IGBC Intent:* Avoid use of refrigerants and ozone depleting gases which has negative impact to the environment.
   - *Approach and Methodologies:* Survey the market for CFC-free HVAC system/ unitary air-conditioners. Install HVAC equipment/ unitary air-conditioners which does not use CFC based refrigerant.
2. *Minimum Energy Performance*
   - *IGBC Intent:* Improve energy efficiency of the building(s) to reduce environmental impacts from excessive energy use.
   - *Approach and Methodologies:* The approach would be to look at orientation, right at the design stage. The design should then consider and select appropriate measures such as insulation of walls, roof, selection of glass, sizing of windows, lighting and efficient air-conditioning & heating systems.

     Identify the materials and equipment available in the market and their properties with regard to energy performance. While selecting these materials and equipment, consider their associated environmental impacts.

     Also, consider the renewable energy feasibility while determining the power requirement from the local utility. The renewable energy sources that can be considered are - Solar energy, Wind energy, Biomass, Bio-gas etc.,
3. *Enhanced Energy Performance*
   - *IGBC Intent:* Enhance energy efficiency of the building(s) to reduce environmental impacts from excessive energy use.
   - *Approach and Methodologies:* The approach would be to look at orientation, right at the design stage. The design should then consider and select appropriate measures such as insulation of walls, roof, selection of glass, sizing of windows, lighting and efficient air-conditioning & heating systems.

     Identify the materials and equipment available in the market and their properties with regard to energy performance. While selecting these materials and equipment, consider their associated environmental impacts.

     Also, consider the renewable energy feasibility while determining the power requirement from the local utility. The renewable energy sources that can be considered are - Solar energy, Wind energy, Biomass, Bio-gas etc.,
4. *On-site Renewable Energy*
   - *IGBC Intent:* Promote self-sufficiency in energy through renewable technologies for on-site power generation and use within the project.
   - *Approach and Methodologies:* Consider renewable energy feasibility while determining the power requirement from the local utility. Renewable energy sources that can be considered are - Solar energy, Wind energy, Biomass, Bio-gas etc.,
5. *Solar Water Heating System*
   - *IGBC Intent:* Encourage use of solar energy for water heating applications in the building(s).

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- Approach and Methodologies: Calculate hot water requirement for the entire project. Install solar water heating system, thereby catering to hot water requirement in the project. The design should also consider the availability of space to install the solar panels.

6. *Energy Saving Measures in Appliances & Other Equipment*
   - *IGBC Intent:* Conserve energy in the use of house-hold appliances and other equipment, thereby reducing environmental impacts.
   - *Approach and methodology:* Conduct market survey before selecting appliances / equipment. Consider the energy efficiency of this appliances / equipment and select minimum BEE 4-star rated or equivalent appliances / equipment.
7. *Distributed Power Generation*
   - *IGBC Intent*: Reduce dependence on fossil fuels for in-situ power generation.
   - *Approach and Methodologies:* Survey the market for available distributed power generators which operate on bio-fuels or non-edible oils, and diesel generator sets which are CPCB compliant for emissions & noise. Also, consider availability of fuel on a sustained basis and economic viability of the system.
8. *Energy Metering*
   - *IGBC Intent:* Encourage continuous monitoring to enhance the energy performance in the project
   - *Approach & Methodologies:* Identify all the major energy consuming equipment and install systems to monitor energy consumption. Develop and implement a measurement & verification mechanism to compare predicted savings to actual energy performance of the installed meters.

2.5 Typical Energy Saving Approach in Buildings

Energy-saving approaches in buildings focus on reducing energy consumption while maintaining thermal comfort, indoor air quality, and occupant productivity. These strategies can be incorporated during the design, construction, and operational stages of a building.

1. Building Orientation - Proper orientation minimizes heat gain in summer and maximizes natural daylight and winter solar heat. Orienting the longer side of the building along the east-west direction reduces direct solar radiation and lowers cooling loads.

2. High-Performance Building Envelope - A well-insulated roof, walls, and floors, along with energy-efficient windows and doors, reduce unwanted heat transfer. Double-glazed windows, insulated walls, and airtight construction help maintain indoor temperatures with less reliance on mechanical heating and cooling.

3. Passive Solar Design - Passive solar techniques utilize natural sunlight for heating and lighting. Features such as shading devices, overhangs, thermal mass, and strategically placed windows reduce artificial energy requirements throughout the year.

4. Natural Ventilation - Cross ventilation, stack ventilation, and well-designed openings improve indoor air circulation and reduce dependence on air conditioning. Proper airflow also enhances indoor comfort and air quality.

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5. Day lighting - Maximizing the use of natural daylight through skylights, clerestory windows, light shelves, and large glazed openings reduces the need for artificial lighting during daytime while improving occupant well-being.

6. Energy-Efficient Lighting - Replacing conventional lamps with LED lighting and installing occupancy sensors, daylight sensors, and automatic lighting controls significantly lowers electricity consumption.

7. Efficient HVAC Systems - High-efficiency heating, ventilation, and air-conditioning (HVAC) systems consume less energy while maintaining thermal comfort. Proper sizing, regular maintenance, and programmable thermostats further improve system efficiency.

8. Renewable Energy Integration - Installing rooftop solar photovoltaic (PV) systems, solar water heaters, and other renewable energy technologies reduces dependence on conventional electricity and lowers greenhouse gas emissions.

9. Energy-Efficient Appliances and Equipment - Using appliances with high energy ratings, variable-speed motors, and efficient pumps and fans minimizes electricity consumption without compromising performance.

10. Building Energy Management Systems (BEMS) - Smart energy management systems monitor and control lighting, HVAC, and other electrical systems in real time. Automated controls optimize energy use based on occupancy and environmental conditions.

11. Water Heating Efficiency - Energy consumption for water heating can be reduced by using solar water heaters, heat pump water heaters, insulated hot-water pipes, and low-flow fixtures.

12. Landscaping and Green Roofs - Trees, vegetation, green roofs, and vertical gardens provide shading, reduce the urban heat island effect, and lower surrounding air temperatures, thereby decreasing cooling energy demand.

13. Regular Maintenance and Energy Audits - Periodic maintenance of building systems and energy audits help identify inefficiencies, equipment faults, and opportunities for energy conservation, ensuring long-term building performance.

14. Occupant Awareness and Energy-Conscious Practices - Educating occupants about energy-saving habits, such as switching off unused lights and equipment, maintaining appropriate thermostat settings, and using daylight effectively, contributes significantly to overall energy savings.

2.6 LEED India Rating System and Energy Efficiency

LEED– Leadership in Energy and Environmental Design

- *Leadership in Energy and Environmental Design (LEED)* is a green building certification program used worldwide.
- Developed by the non-profit U.S. Green Building Council (USGBC), it includes a set of rating systems for the design, construction, operation, and maintenance of green buildings,

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homes, and neighbourhoods, which aims to help building owners and operators be environmentally responsible and use resources efficiently.

- According to LEED itself, "The LEED plaque on a building is a mark of quality and achievement in green building."
- As such, any building with LEED certification has proven itself to be designed, built, and maintained with the highest degree of energy efficiency.
- There are a number of benefits of LEED certification such as:
    1. Lower operating costs;
    2. Reduction in risks and liability;
    3. More durable buildings;
    4. Better indoor air quality;
    5. Improved resale value; and
    6. Support for the local economy.
- LEED projects are typically scored across nine areas that each speaks to a specific aspect of green building design:
    1. Integrative process;
    2. Location and transportation;
    3. Sustainable sites;
    4. Water efficiency;
    5. Energy and atmosphere;
    6. Materials and resources;
    7. Indoor environmental quality;
    8. Innovation; and
    9. Regional Priority.
- This rating system applies to new construction projects, existing buildings, neighborhood developments, and even cities and communities.
- Each certification will be valid for three years, at the end of which the project will need to get recertified again.
- If your project qualifies for a LEED certificate, it will earn one of the following four ratings:
    i. Platinum: 80-plus points earned;
    ii. Gold: 60-79 points earned;
    iii. Silver: 50-59 points earned; or
    iv. Certified: 40-49 points earned.
- This rating system applies to new construction projects, existing buildings, neighborhood developments, and even cities and communities.

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